

WIPRO CAPSTONE PROJECT

Test Analysis Report

BlazeDemo Flight Booking Automation Framework

Selenium WebDriver · TestNG · Maven · ExtentReports · Jenkins · Docker

Project Name	BlazeDemo Flight Booking Automation Framework
Document Type	Test Analysis Report
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Application Under Test	BlazeDemo
URL	https://blazedemo.com/
Positive Suite Result	PASS: 12/12 (100%)
Negative Suite Result	PASS: 3/3 (100%)
Defects Logged	7 (BLAZE-7 to BLAZE-13)
Repository	ThisIs-Developer/Wipro-Capstone-Project
GitHub Link	https://github.com/ThisIs-Developer/Wipro-Capstone-Project
Date	13 th June 2026

1. Executive Summary

This Test Analysis Report summarizes the results of testing conducted on the BlazeDemo Flight Booking web application. Testing was performed using Selenium WebDriver with the Page Object Model (POM) framework and TestNG. Two test suites were executed: a positive suite that tested the complete flight booking process with three sets of passenger data, and a negative test suite that checked the system's response to invalid inputs and Authentication module.

All test cases passed successfully, showing that the main booking process works as expected. However, 7 defects were identified and logged in JIRA.

2. Scope of Testing

The following modules were included in the testing scope:

- Home Page
- Flight Search
- Flight Selection
- Purchase Page
- Booking Confirmation
- Registration Page
- Login Page

The testing focused on validating critical business workflows and identifying application defects.

3. Testing Types Performed

- **Functional Testing:** Validated application functionality against expected behavior.
- **End-to-End Testing:** Verified the complete flight booking workflow from flight search to booking confirmation.
- **Data-Driven Testing:** Executed booking scenarios using multiple passenger datasets from Excel.
- **Negative Testing:** Validated application behavior using invalid, blank, and malformed inputs.
- **Smoke Testing:** Verified critical application functionalities before detailed execution.
- **Regression Testing:** Ensured framework stability across multiple executions.
- **Cross-Browser Testing:** Verified application functionality across different web browsers to ensure consistent behavior.

4. Overall Execution Summary

Metric	Result
Total Test Cases Executed	12(4 Steps × 3 Passengers)
Passed	12
Failed	0
Skipped	0
Success Rate	100%
Failure Rate	0%

5. Suite-Level Test Execution Results

Test Suite	Suite File	Tests	Passed	Failed	Skipped	Result
FlightBookingSuite	testng.xml	12	12	0	0	PASS

6. Module Execution Analysis

Application Module	Validation Performed	Status
Home Page	URL, title, city selection	Passed
Flight Search	Flight availability and search results	Passed
Flight Selection	Flight selection process	Passed
Purchase Page	Passenger and payment information	Passed
Booking Confirmation	Booking details verification	Passed

7. Functional Validation Summary

- Home page validation
- Flight search functionality
- Flight selection process
- Passenger information submission
- Payment information submission
- Booking completion workflow
- Booking confirmation verification

8. Test Coverage Overview

Testing Area	Coverage
Functional Testing	100%
End-to-End Testing	100%
Data-Driven Testing	100%
Smoke Testing	100%
Regression Testing	100%
Cross-Browser Testing	Executed

9. Execution Environment & Browser Compatibility

- **Local Execution:** Chrome Browser (UI Mode) executed successfully through Eclipse IDE and TestNG.
- **Maven Execution:** Test suite executed successfully using `mvn clean test`.
- **CI/CD Execution:** Automated test execution completed successfully through Jenkins pipeline.
- **Docker Execution:** Framework executed successfully inside Docker containerized environment.
- **Browser Support:** Framework supports execution on Chrome, Firefox, and Edge browsers through configurable settings.

Run as TestNG Suite in IDE

Console

```
:terminated> BlazeDemo_testng.xml [TestNG] C:\Users\User\p2\pool\plugins\org.eclipse.justj.openj
Jun 12, 2026 12:18:28 PM org.openqa.selenium.devtools.CdpVersionFinder fin
WARNING: Unable to find an exact match for CDP version 149, returning the
2026-06-12 12:18:28 [main] INFO Log - Launching Browser : chrome
2026-06-12 12:18:30 [main] INFO Log - Application Opened
STARTED : verifyHomePage
2026-06-12 12:18:30 [main] INFO Log - Home Page Verification Started
2026-06-12 12:18:31 [main] INFO Log - Home Page Verification Completed
PASSED : verifyHomePage
STARTED : verifyFlightSearch
2026-06-12 12:18:31 [main] INFO Log - Flight Search Verification Started
2026-06-12 12:18:32 [main] INFO Log - Flight Search Verification Complete
PASSED : verifyFlightSearch
STARTED : verifyPurchasePage
2026-06-12 12:18:32 [main] INFO Log - Purchase Page Verification Started
2026-06-12 12:18:34 [main] INFO Log - Purchase Page Verification Complete
PASSED : verifyPurchasePage
STARTED : verifyBookingConfirmation
2026-06-12 12:18:34 [main] INFO Log - Confirmation Verification Started
2026-06-12 12:18:34 [main] INFO Log - Screenshot Captured
2026-06-12 12:18:34 [main] INFO Log - Booking Confirmation Completed
PASSED : verifyBookingConfirmation
2026-06-12 12:18:34 [main] INFO Log - Closing Browser

=====
FlightBookingSuite
Total tests run: 12, Passes: 12, Failures: 0, Skips: 0
=====
```

Results of running suite

Search: Passed: 12 Failed: 0 Skipped: 0 1/1 Methods: 12 (37997 ms)

All Tests Failed Tests Summary

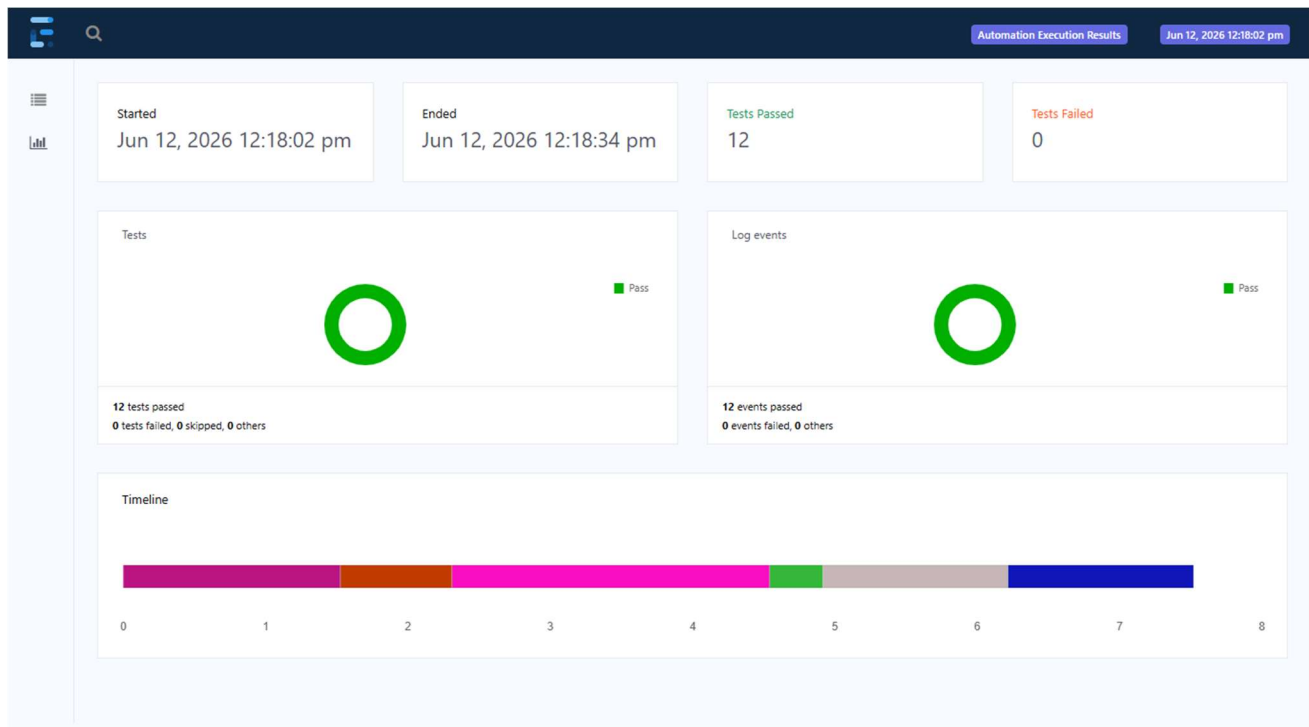
- FlightBookingSuite (12/0/0/0) (15.332 s)
 - FlightBookingTests (15.332 s)
 - testcases.FlightBookingTest
 - verifyHomePage (1.563 s)
 - verifyFlightSearch (0.409 s)
 - verifyPurchasePage (2.825 s)
 - verifyBookingConfirmation (1.059 s)
 - verifyHomePage (1.308 s)
 - verifyFlightSearch (0.779 s)
 - verifyPurchasePage (2.172 s)
 - verifyBookingConfirmation (0.428 s)
 - verifyHomePage (1.313 s)
 - verifyFlightSearch (0.801 s)
 - verifyPurchasePage (2.239 s)
 - verifyBookingConfirmation (0.436 s)

Mvn clean test

```
2026-06-13 10:37:43 [main] INFO Log - Home Page Verification Completed
PASSED : verifyHomePage
STARTED : verifyFlightSearch
2026-06-13 10:37:43 [main] INFO Log - Flight Search Verification Started
2026-06-13 10:37:44 [main] INFO Log - Flight Search Verification Completed
PASSED : verifyFlightSearch
STARTED : verifyPurchasePage
2026-06-13 10:37:44 [main] INFO Log - Purchase Page Verification Started
2026-06-13 10:37:47 [main] INFO Log - Purchase Page Verification Completed
PASSED : verifyPurchasePage
STARTED : verifyBookingConfirmation
2026-06-13 10:37:47 [main] INFO Log - Confirmation Verification Started
2026-06-13 10:37:48 [main] INFO Log - Screenshot Captured
2026-06-13 10:37:48 [main] INFO Log - Booking Confirmation Completed
PASSED : verifyBookingConfirmation
2026-06-13 10:37:48 [main] INFO Log - Closing Browser
[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 52.65 s -- in TestSuite
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:03 min
[INFO] Finished at: 2026-06-13T10:37:49+05:30
[INFO] -----
```

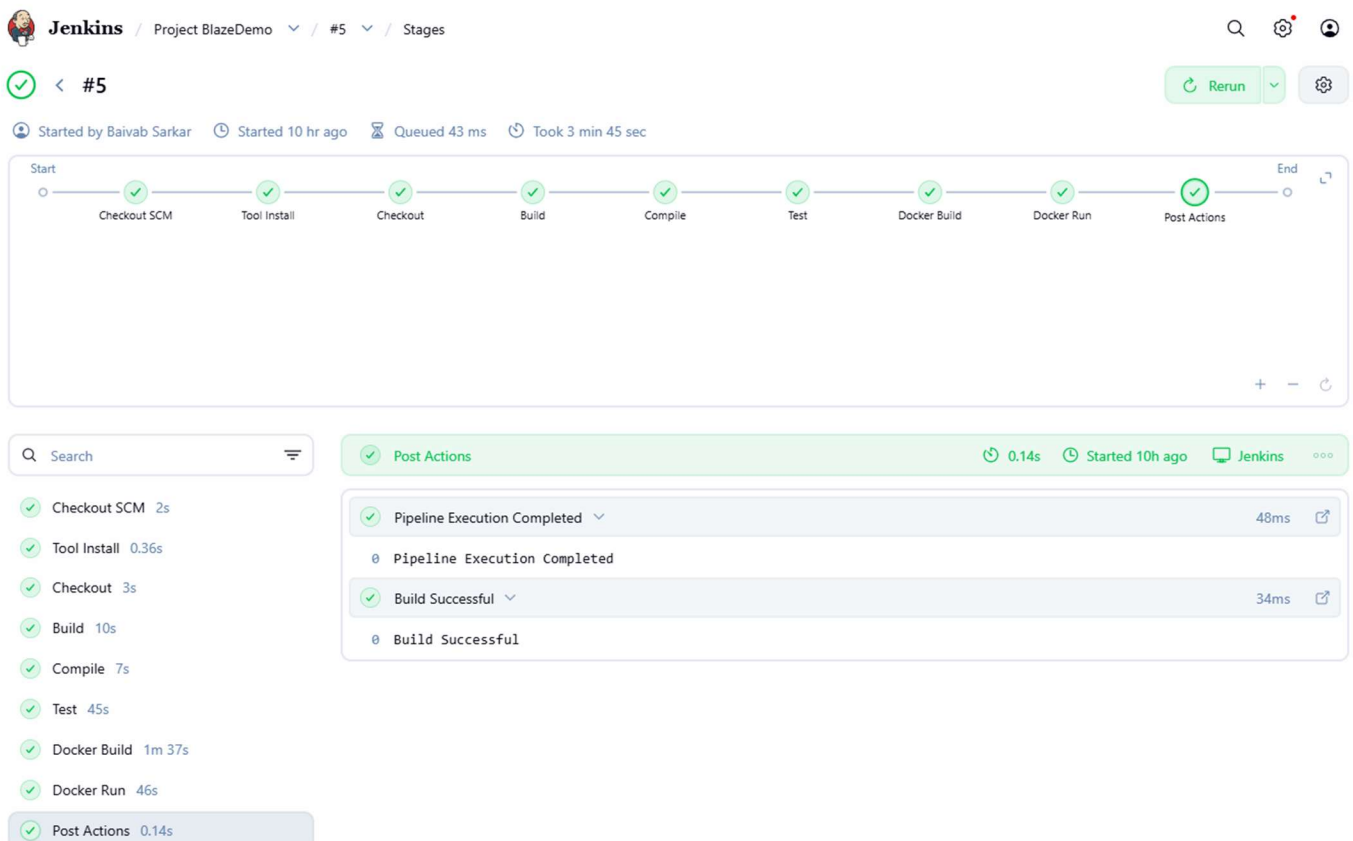
10. Extent Report Analysis

The Extent Report provides a detailed summary of the BlazeDemo Automation Framework execution, including test status, execution timestamps, screenshots, and execution logs. During execution, the data-driven framework processed three passenger datasets and executed 12 test methods. The report recorded a 100% pass rate with no failed or skipped test cases. Screenshots and execution logs were attached as supporting evidence for test validation.



11. Jenkins Pipeline Analysis

The Jenkins pipeline successfully automated the build and test execution process for the BlazeDemo Flight Booking Automation Framework. The pipeline executed all configured stages, including source code checkout from GitHub, Maven build, project compilation, test execution, Docker image creation, and container execution. All stages completed successfully without failures, demonstrating the stability of the CI/CD integration.



12. Docker Execution Analysis

The automation framework was successfully containerized using Docker and executed within a Selenium Chrome standalone environment. The Docker image was built using the project Dockerfile,

The first screenshot shows the Docker Desktop interface with the 'Logs' tab selected for the container 'stoic_mclean'. The logs display a successful test execution with the following details:

- Container ID: a84d1e26f28d
- Image: blazedemo-framework:latest
- Status: Exited (0) (4 minutes ago)
- Log messages include: PASSED : verifyPurchasePage, STARTED : verifyBookingConfirmation, 2026-06-12 07:00:32 [main] INFO Log - Confirmation Verification Started, 2026-06-12 07:00:32 [main] INFO Log - Screenshot Captured, 2026-06-12 07:00:32 [main] INFO Log - Booking Confirmation Completed, PASSED : verifyBookingConfirmation, 2026-06-12 07:00:32 [main] INFO Log - Closing Browser, [INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 39.87 s -- in TestSuite, [INFO] Results:, [INFO] BUILD SUCCESS, [INFO] Total time: 48.669 s, [INFO] Finished at: 2026-06-12T07:00:33Z.

The second screenshot shows the 'Logs' view in Docker Desktop, displaying a table of log entries for the same container. The table has three columns: Timestamp, Object, and Message. The log entries are as follows:

Timestamp	Object	Message
2026-06-12 12:30:32	stoic_mclean	2026-06-12 07:00:32 [main] INFO Log - Screenshot Captured
2026-06-12 12:30:32	stoic_mclean	2026-06-12 07:00:32 [main] INFO Log - Booking Confirmation Completed
2026-06-12 12:30:33	stoic_mclean	PASSED : verifyBookingConfirmation
2026-06-12 12:30:33	stoic_mclean	2026-06-12 07:00:32 [main] INFO Log - Closing Browser
2026-06-12 12:30:33	stoic_mclean	[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: ...
2026-06-12 12:30:33	stoic_mclean	[INFO]
2026-06-12 12:30:33	stoic_mclean	[INFO] Results:
2026-06-12 12:30:33	stoic_mclean	[INFO]
2026-06-12 12:30:33	stoic_mclean	[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0
2026-06-12 12:30:33	stoic_mclean	[INFO]
2026-06-12 12:30:33	stoic_mclean	[INFO] BUILD SUCCESS
2026-06-12 12:30:33	stoic_mclean	[INFO]
2026-06-12 12:30:33	stoic_mclean	[INFO] Total time: 48.669 s
2026-06-12 12:30:33	stoic_mclean	[INFO] Finished at: 2026-06-12T07:00:33Z
2026-06-12 12:30:33	stoic_mclean	[INFO]

5. Conclusion

The BlazeDemo Flight Booking Automation Framework successfully validated the end-to-end flight booking process and achieved a 100% test execution success rate. The framework demonstrated stable execution with Selenium WebDriver, TestNG, Jenkins, and Docker, confirming the effectiveness of the automation solution and successful completion of the project objectives.